



Increasing incidence of anterior cruciate ligament reconstruction: a 17-year population-based study

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Abstract

Purpose Anterior cruciate ligament (ACL) injuries are one of the most frequently studied injuries in orthopedic care and research. However, limited epidemiological data are available in Canada regarding trend and distribution of anterior cruciate ligament reconstruction (ACLR). In this paper, our purpose was to assess trends of ACLR between 2002/03 and 2018/19 by age, sex, season of surgery, and location (inpatient vs outpatient) of surgery.

Methods In this descriptive epidemiological study of retrospective data available from Alberta Ministry of Health, we report annual incidence of ACLR between 2002/03 until 2018/19 among Albertans aged 10 years and older. Information was collected by authors from physician claims database for primary ACLR and revision ACLR and linked with other databases. Incidence proportions (number of ACLR/100,000 population) were calculated and compared by age category and gender over the study period.

Results A total of 28,401 primary ACLR and 2085 revision ACLR were identified during the study period. Age-standardized annual incidence of primary ACLR increased from 40.6 to 51.2 per 100,000 population aged 10 years and older. Average annual increase in ACLR incidence was higher among females (1.8% per years) compared to males (0.96% per year). The overall peak incidence and peak incidence among males was observed in 20–29 year age group, whereas peak incidence in females was observed in 10–19 years of age. The number of ACLR in females outnumbers those among males for 10–19 year age group. Generally, a lower proportion of ACLR were conducted in summer compared to other seasons. Primary ACLR conducted in outpatient setting increased from 72% in 2002/03 to 97% in 2018/19.

Conclusion The incidence of ACLR is increasing in Alberta, especially among females and among younger cohorts under 20 years of age. This information can help clinicians to provide patient education and policy-makers to design and implement targeted ACL injury prevention programs.

Level of evidence Level III.

Keywords Anterior cruciate ligament reconstruction · Trends · Epidemiology · Cruciate ligament injury

Abbreviations

ACL	Anterior cruciate ligament
ACLR	Anterior cruciate ligament reconstruction
NACRS	National ambulatory care reporting system
DAD	Discharge abstract database

Introduction

Anterior Cruciate Ligament (ACL) rupture is a commonly reported injury among younger athletes [2, 21]. Over a period of 25 years, one in 29 female athletes and one in 50 male athletes are reported to experience an ACL injury [31]. Recent studies indicate that, although ACL injury rate in general population of males has declined over time, it has remained unchanged among females [36, 41].

There is an ongoing debate about the optimal management of ACL injury in general population [15]. Some studies suggest that anterior cruciate ligament reconstruction (ACLR), when indicated, is a cost-effective treatment for ACL injury [12, 29]. Whereas other studies show that patient-reported outcomes at 2 and 5 years did not differ

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between conservative management and surgical reconstruction group [30, 32]. Surgical reconstruction is generally recommended for patients who are young and had high pre-injury activity level [20], as well as who want to return to usual activity especially for those in organized sports or those in physically demanding occupations [4, 8]. Furthermore, patients with signs of recurring instability are often offered ACLR to restore stability and protect meniscus and cartilage tissue [4]. Literature suggests that timely diagnosis and optimal management of ACL injury can prevent intra-articular injuries by reducing chances of instability episodes [43]. Recent studies showed that delaying surgery beyond 6 months was associated with increased risk of revision surgery, and reduced chances to achieve clinically significant outcomes [18].

ACLR incidence is found to be increasing in other jurisdictions [45, 51]. However, limited data are available in Alberta examining the trend of primary and revision ACLR. A seasonal pattern of ACL injury and effect of season on surface condition was reported in some studies [1, 35]; however, data on seasonal pattern of ACLR in Alberta are currently unavailable. Therefore, this study was conducted to estimate trends of ACLR from 2002/03 to 2018/19 by age, sex, outpatient/inpatient setting, and season. Knowledge of the disease burden is important for proper allocation of scarce resources. More importantly, information on the most at-risk groups will be helpful to prioritize populations for safety promotion and injury prevention. As well, this information can be used by sports practitioners and policy-makers to design and implement tailored prevention programs and by clinicians to prioritize populations for patient education about the risk of ACL injury. We hypothesized that both primary ACLR and revision ACLR are increasing over time.

Materials and methods

Ethical approval for this study was obtained from the University of Alberta Human Research Committee (Pro00090820). In this descriptive epidemiological study, administrative databases available from Alberta Ministry of Health were used. Primary ACLRs and revisions conducted between 2002/03 to 2018/19 were identified using physician claims database. Physician claims database was linked with the national ambulatory care reporting system (NACRS) and the inpatient discharge abstract database (DAD) to identify reconstruction setting.

Exclusion criteria

Primary and revision ACLR conducted between 2002/03 and 2018/19 in Alberta were included in the study. Early

ACLR repairs (less than 14 days) and ACLRs conducted in Non-Albertans and among those with missing demographic data were excluded. Patients below the age of 10 years were excluded due to sparse ACLR data in this age group. Duplicate records of ACLR conducted on a person on the same date were excluded. Records of more than two primary ACLRs on a person were excluded from analysis. The number of patients excluded for different reasons is shown in Fig. 1.

Measures

The outcome measures for this study were primary ACLR and revision ACLR. The distribution of ACLR by age, sex, setting of surgery, and season of surgery was assessed by the authors. Age groups were categorized into 10–19, 20–29, 30–39, 40–49, 50–59, and 60 years and above. Sex was categorized into males and females. Surgery setting was classified into outpatient and inpatients. Season of ACLR was divided into spring, summer, fall, and winter.

Statistical analysis

Age-standardized ACLR rates were calculated for the total population and by sex considering Canadian population 2011 as a standard population. The average annual growth rate of primary ACLR was compared between males and females by age group. The proportion of ACLR was calculated using the number of primary/revision ACLR in each fiscal year as the numerator. The population of specific age and sex groupings reported in the closest census was used as the denominator (Additional file 1).

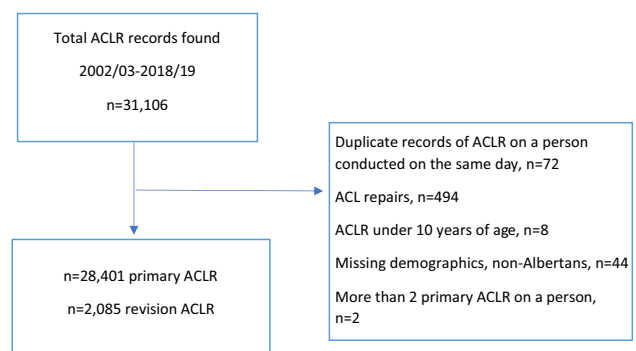


Fig. 1 Sample selection flowchart

Results

Primary ACL reconstructions

From 2002/03 until 2018/19, there were a total of 28,401 primary ACLR performed in Alberta in patients aged 10 years and older (Additional file 2). Of them 27,124 (95.5%) were initial primary ACLR and 1274 (4.5%) were contralateral primary ACLR. Of the total primary ACLRs, 56% were performed on males and 44% on females.

The annual number of primary ACLRs increased from 1137 in 2002/03 to 1894 in 2018/19 (Supplementary Table A2). The annual age-standardized incidence increased from 40.6 per 100,000 population in 2002/03 to 51.2 per 100,000 population in 2018/19, an increase of 26% over the study period with an annual growth rate of 1.3% (Fig. 2). Among males, ACLR incidence increased

from 44.7 per 100,000 population to 54.9 per 100,000 population (Fig. 2). Among females, it increased from 32.9 per 100,000 to 47.5 per 100,000 population.

The highest incidence was reported among males in the 20–29-year age group (Fig. 3). The peak incidence among males was reported in 2015/16 among the 20–29-year age group (147.2/100,000 population). Females had the highest incidence among 10–19-year age group (Fig. 4). The peak incidence among females was reported in 2017/18 (108.8/100,000 population) among in the 10–19-year age group. ACLR rates were highest among 10–19-year-old females compared to male counterparts throughout the study period.

Annual growth rates of primary ACLR were higher among adolescents (10–19 years old), and older age categories (40–49 years old and 50 years old and above) compared to middle age groups among both males and females (Fig. 5). The incidence rate of primary ACLR

Fig. 2 Annual age-standardized incidence of primary ACLR per 100,000 population among males and females aged 10 years and above (standard population: 2011 Canada population)

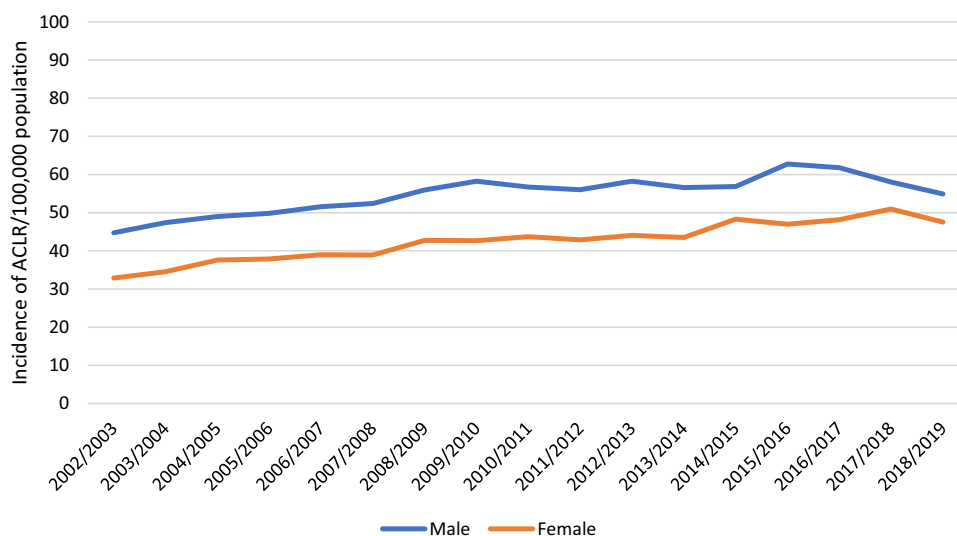


Fig. 3 Incidence trend of primary ACLR per 100,000 population by age groups in males

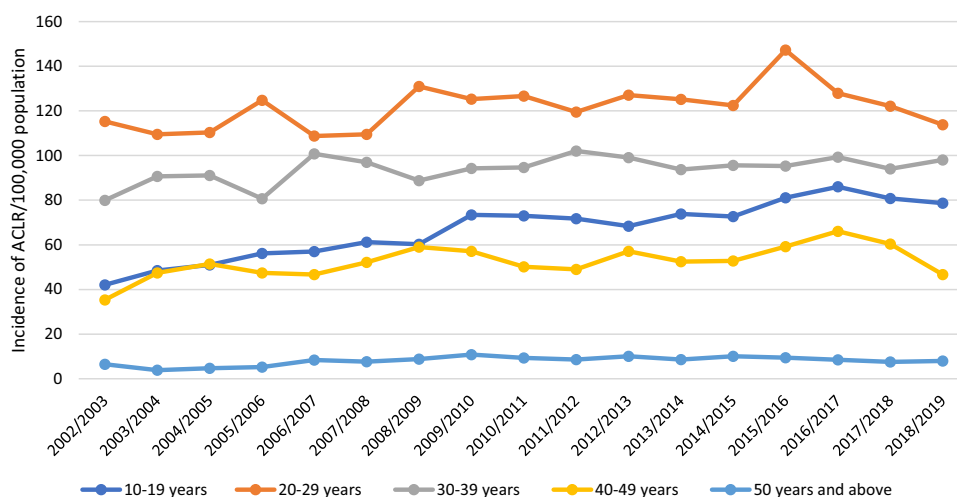


Fig. 4 Incidence trend of primary ACLR per 100,000 population by age groups in females

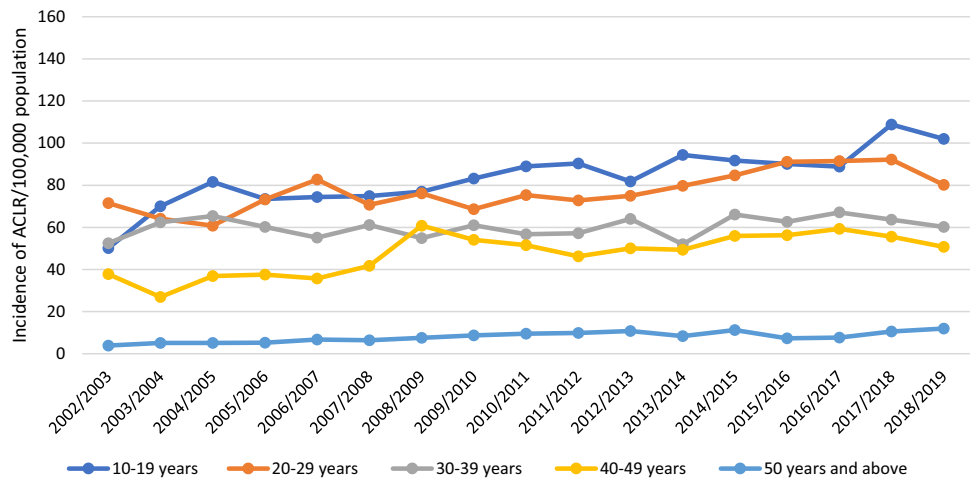
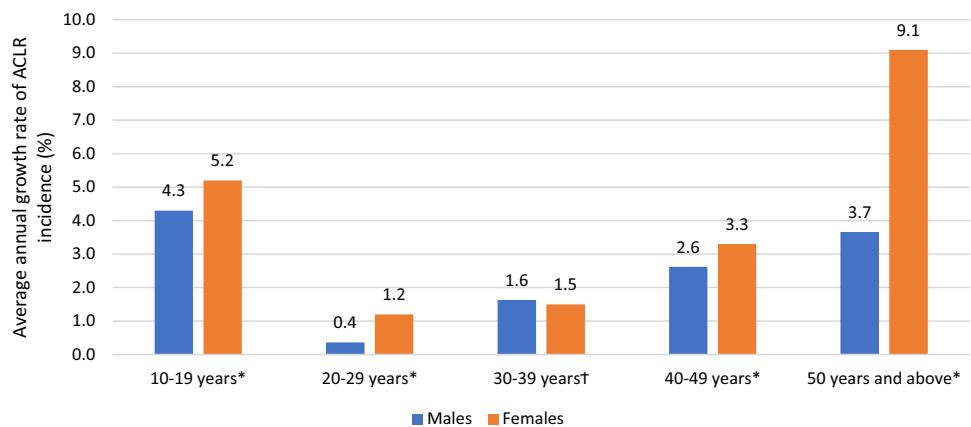


Fig. 5 Annual growth rate in primary ACLR by age groups and sex. * $p < 0.0001$, † $p = 0.07$



increased from 3.9 to 12/100,000 population among females aged 50 years and above with an annual growth rate of 9.1% per year. Similarly, the rate increased from 50.1 to 102.0/100,000 population among females aged 10–19 years. Among males, the highest annual growth rate was reported among those 10–19 years old (4.3%), followed by 50 years old and above (3.7%) and 40–49 years old (2.6%).

There was a strong trend toward outpatient (same-day) surgery over the course of the study. The proportion of primary ACLR conducted in an outpatient setting/same-day surgery increased from 72% in 2002/03 to 97% in 2018/19. The highest increase in proportion of surgeries conducted in an outpatient setting compared to previous years was observed in 2007/08 (7%) and 2014/15 (12%).

Similar proportions of primary ACLRs were performed in spring, fall, and winter seasons and slightly lower proportions were conducted in summer. In 2018/19, of the 1894 primary ACLRs, 509 (26.9%) were conducted in spring, 487 (25.7%) in fall, and 492 (26%) in winter, whereas 406 (21.4%) were conducted in summer.

Revision ACL reconstructions

A total of 2085 ACLR revisions were reported in this period. Of them, 1094 (52.5%) were among males and 991 (47.5%) were among females. The annual number of revision ACLRs increased from 53 in 2002/03 to 195 in 2018/19. The annual incidence of ACLR revision increased from 2.1 per 100,000 population to 5.5 per 100,000 population, an increase of 168% over the study period. Age-standardized annual rates of revision ACLRs increased from 1.9 per 100,000 to 5.4 per 100,000 among males. For females, age-standardized annual rates of ACLR increased from 1.8 per 100,000 to 5.0 per 100,000 population.

Of the 2085 revisions, 1940 (93.1%) were first revisions and 145 (6.9%) were either repeat revisions or revisions on a contralateral knee. First revisions were reported in ($n = 1940$, 6.8%) of total primary ACLR ($n = 28,401$) conducted between 2002/03 and 2018/19. However, some of the revision cases observed during the study period may be among individuals who underwent primary ACLR before 2002/03.

Discussion

The most important finding from this study was that the annual incidence of primary ACLR was found to be increasing in Alberta with an average annual growth rate of 1.3%. The annual age-standardized incidence increased by 26% over the study period. This finding is consistent with findings from studies conducted in other settings [9, 45, 51]. The increase in ACLR may be partly due to growing incidence of ACL injury due to increased sports participation or intention to remain active among a wider age range of population. It may also have been driven by improved surgical technique and increased comfort with the ACLR procedure.

The next important finding from this study was that adolescents (10–19 years old) and adults (50 years old and above) showed a higher annual increase compared to other age categories. Specially, females aged 10–19 years and 50–59 years reported the highest annual increase compared to other age categories. Research from the United States and Australia also showed an increasing ACLR trend in adolescents [10, 46, 51]. Tepolt et al. using data from 45 hospitals in the United States reported a 2.8 times increase in ACLR in the pediatric population, especially in 11–18 years old, compared to total orthopedic operations in the same population over a 10-year period between 2004 and 2014 [46]. Studies from the United States and Sweden have shown that female adolescents have a higher ACL injury risk compared to males involved in soccer, basketball, and softball [42, 48]. The higher annual increase among adolescent females may be due to increased popularity of multidirectional sports, such as soccer and basketball. Additionally, current preferred practice among surgeons is to recommend surgical reconstruction among the active pediatric population with complete ACL tears since technical advances have improved outcomes in skeletally immature patients [13, 38]. Furthermore, new evidence showing superior outcomes of ACLR over non-operative management with minimal growth disturbance [24, 47], and adolescents reaching skeletal maturity earlier than previously thought may also play a role [6]. Increases among the adult population (50 years old and above) may be due to efforts to stay active during adulthood or increased comfort and satisfaction with ACLR [40] due to advances in surgical technique and instruments.

The timing and intricacies of surgical reconstruction among skeletally immature patients are still a contentious issue. It is suggested that early surgical intervention is an appropriate strategy compared to delayed reconstruction or non-operative management [13, 14, 39]. However, high risk of re-rupture and some concern for growth disturbance resulting in leg length discrepancy remains prominent [23, 26, 27] despite technological advances.

In addition to impacts on immediate physical and psychosocial health, ACL injury among adolescents increases risk of early onset osteoarthritis [44]. The odds of total knee replacement in adults with ACL injury is seven times higher in comparison to those without ACL injury [22], and the risk increases with early onset osteoarthritis. A systematic review and metaanalysis revealed that only 65% of athletes with an ACL injury returned to pre-injury level of sport [4]. Remaining inactive after ACL injury increases the chances of adiposity [28]. The chances of returning to pre-injury activity level may be much lower in the general population. The chances of returning to pre-injury level of activity are nearly half among females compared to males [3]. Therefore, the growing incidence of ACL injury in the young population is worrying, because future burden of degenerative disease will be heavily borne by today's younger cohorts. Additionally, it adds strain to the health system now and in the future [17].

Current analysis reveals that the incidence of ACLR peaked in males and females in different age groups. Among males, ACLR incidence was the highest among 20–29 years old throughout the study period. Among females, peak incidence rate was observed between 10 and 19 years old for most of the study period. While the population of adolescent females was less than adolescent males in the province, the absolute number of ACLR performed in adolescent females (10–19 years) was greater than in adolescent males for most years. This finding might reflect higher number of ACL injuries among adolescent females compared to males associated with increasing participation of high school females in competitive sports [41]. A recent metaanalysis found that adolescent girls were at about 1.5 times greater risk of ACL injury compared to males with females showing four times greater risk for basketball compared to males [7]. Additionally, a study from the United States showed that the risk of secondary ACL injury—mainly contralateral ACL injury—was higher among adolescent females than males [37]. Furthermore, it could also be due to female adolescents attaining skeletal maturity 1–2 years earlier than male adolescents [33]; and higher preference for surgical reconstruction among female adolescents.

However, overall ACLR incidence proportion was higher in males (57.9/100,000 population) compared to females (49.4 per 100,000 population) in 2018/19. Overall, incidence of ACLR was reported higher in males compared to females in other settings, as well [41, 45]. In our study, males reported the highest incidence between 20 and 29 years of age (113.9 per 100,000 population). The higher proportion among males may be due to a higher proportion of males remaining active compared to females and may also be due to participation bias with males more likely to be participating in higher-risk sports than females. However, our results indicate that the gender gap in ACLR incidence

has narrowed over time as a consequence of greater increase in ACLR incidence in females compared to males. This might reflect a decline in ACL injury incidence in males. In a 21-year population-based study from USA, Sanders et al. found that ACL injury incidence has declined in males, whereas it has remained stable among females [41].

Consistent with findings from an Australian study [51], we found a higher increase in annual incidence of revision ACLR compared to primary ACLR. One of the reasons for higher revision ACLR may be due to growing number of ACLR being performed in younger age groups (10–19 years), who are more likely to return to pre-injury activity level earlier compared to older age groups [49]. Additionally, graft selection, tunnel placement [25], or other factors may also have played a role.

Growing evidence shows that implementing neuromuscular training programs can effectively reduce ACL injury. A meta-analysis of metanalysis showed that injury prevention programs can reduce ACL injury by 50% in all athletes and non-contact ACL injuries by 67% among female athletes [50]. Translating this evidence from RCTs and metanalyses to large-scale implementation is the current challenge for stakeholders [11, 19]. Identifying, context-specific barriers and facilitators so as to design best strategies to facilitate widespread adoption and sustainable implementation are now needed [16, 34]. The benefits of the growing number of females participating in sports will truly be reaped in terms of long-term health outcomes if such injury prevention programs can be implemented on a mass scale on a routine basis.

Consistent with recent trends in other settings [5], there was a significant drop in proportion of ACLR conducted in inpatient settings with a simultaneous increase in proportion of ACLRs conducted in an outpatient setting. Advances in surgical and anesthesia techniques and equipment have made ACLR a simple surgical procedure that can be successfully performed in an outpatient setting. Furthermore, our study found that lower proportion of ACLR was performed in summer compared to other seasons. This pattern may be associated with intention to return to normal activity by reconstructing torn ACL before summer arrives or may be due to constrained staff availability in summer. However, this needs further investigation.

This is one of the few studies examining the recent trends of primary and revision ACLR using population-based administrative data. The increasing incidence of ACLR found in the current study, comparable to reports from other jurisdictions in Canada [52], and elsewhere [45, 51] may be generalized to other settings with similar health systems and similar rates of participation in sports and recreational activities. However, the study has some limitations. Since we do not have data on the number of patients undergoing non-surgical management, the incidence of ACL injury is

likely higher than incidence of ACLR. However, the growing ACLR incidence might reflect growing ACL injury incidence or a change in treatment pattern. Physician billing codes may change over time or may have been miscoded. However, we verified our primary ACLR records from physician billing codes with NACRS and inpatients databases associated with ligament reconstruction, and we found a match of 96%. Therefore, we assume that physician billing codes are a reliable source for ACLR data.

Conclusions

The incidence of primary ACLR is increasing in Alberta especially in adolescents, which most likely reflects a growing incidence of ACL injury in the province. Since ACL injury at a young age increases the risk of early onset osteoarthritis at a young age, implementing ACL injury prevention programs with a special focus on adolescent population is critical.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00167-022-07093-1>.

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Author contributions This study is a part of first author's doctoral thesis. All authors contributed substantially to the conception and design of the study. DV and YRP were responsible for acquisition of data. YRP conducted the analysis and wrote the original draft of manuscript. MS and DV contributed to review, and interpretation of data. All authors approved the final version of this manuscript.

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Declarations

Conflict of interest We declare that there is no conflict of interest.

Ethical approval This study was approved by the University of Alberta Human Research Committee (Pro00090820).

Informed consent Since we used deidentified patient records from Alberta Ministry of Health, informed consent was not required.

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